F1 Exploratory Data Analysis

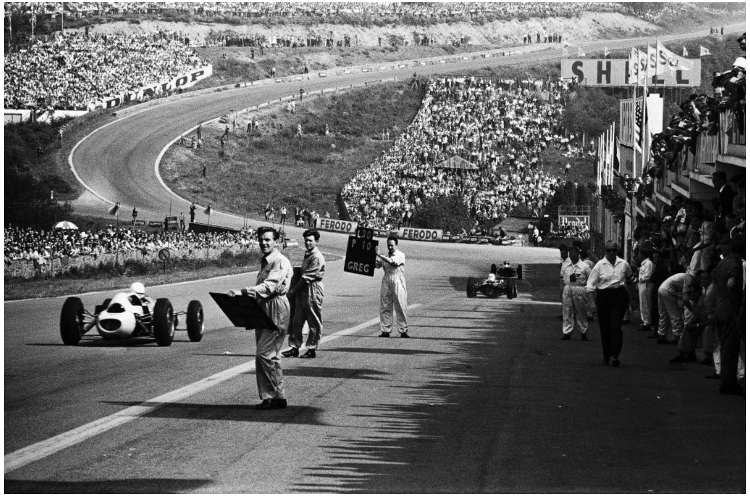
Avi Khurana, Christian Haranzo, Teresa Bui, Yernar Smagulov

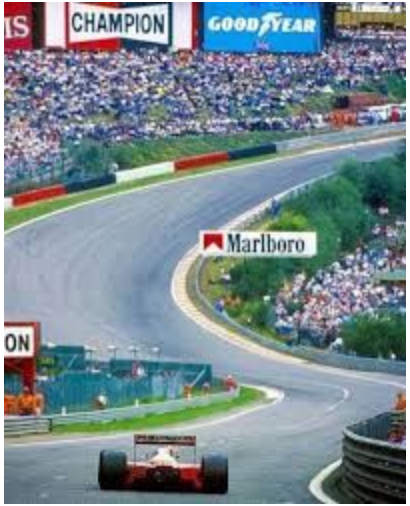
Background & Purpose

- Explore F1 data over time to analyze performance trends across eras
- F1 technology and rules have evolved over time
    - This analysis will look into how these changes link to the performance of drivers and teams

Content

- Analysis uses datasets covering results from 1950s to modern day
- Data contains results by driver and team
    - By season, by era, and by career





Guiding Questions

1. How has F1 constructor and driver performance changed over time measured via various performance metrics including race wins, race start percentage, championships won, and more?

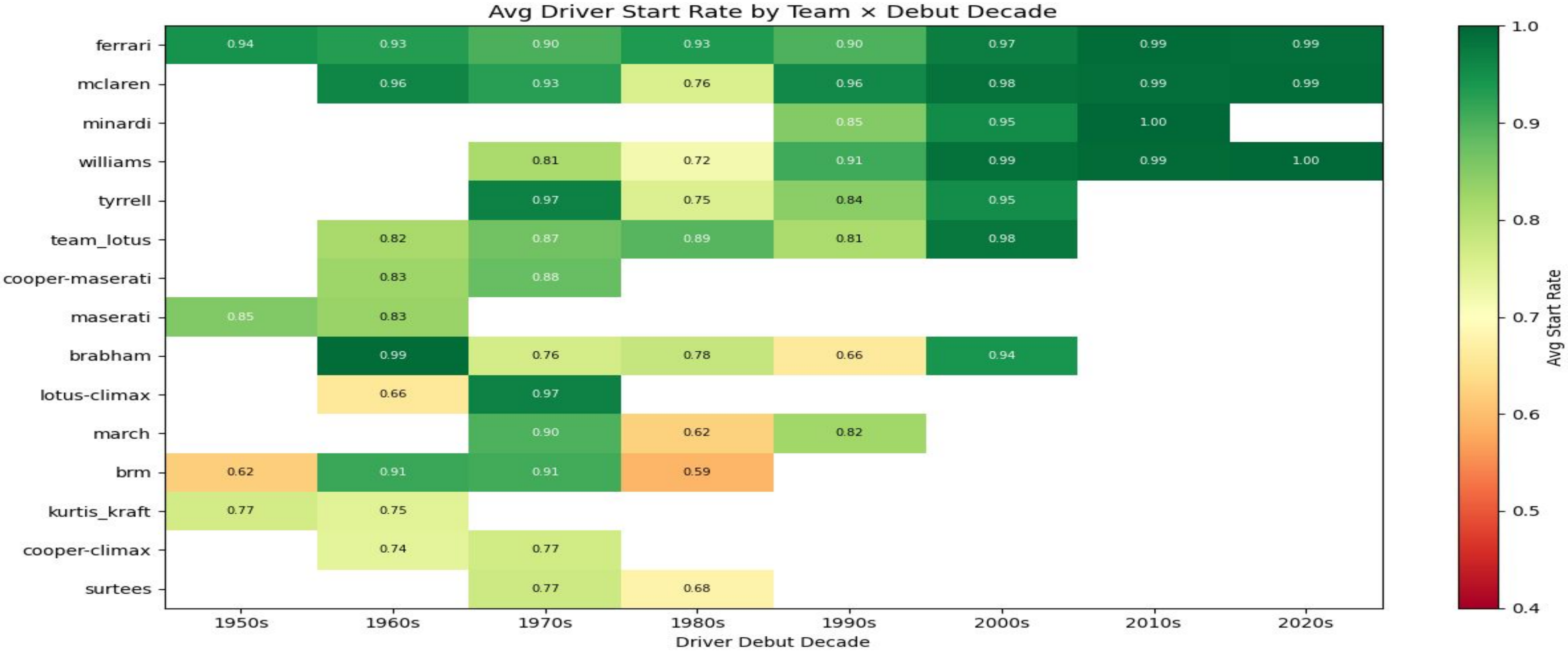
2. How do the career performance metrics of the best drivers compare with one another?

Data Preparation & Cleaning

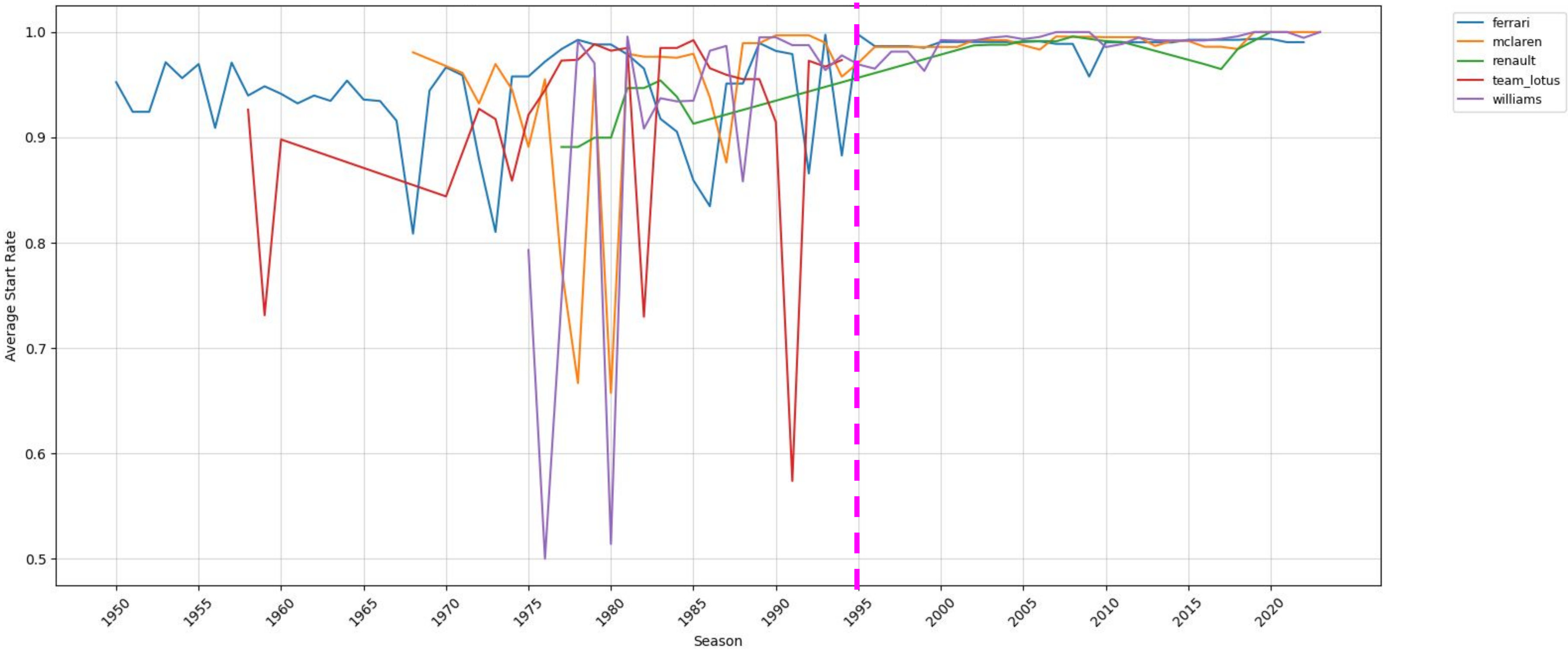
- **7 Data Sources:** Career driver data, races, results, constructors, constructor standings, driver standings, qualifying

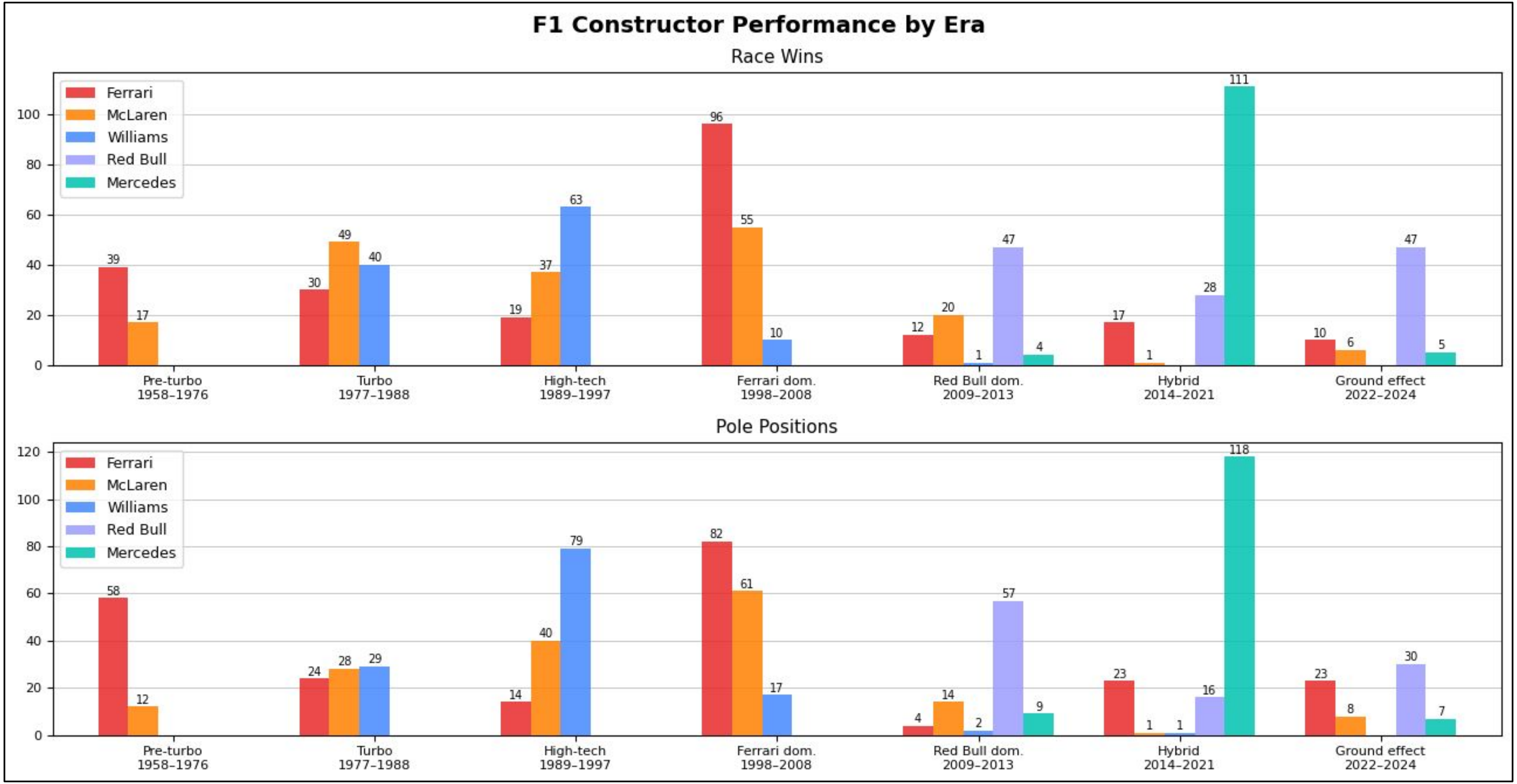
- **Key Challenges & Solutions:**
  - String-to-list parsing: Championship Years and Seasons cols stored as strings; explode to create one row per season
  - ID-to-name translation: Raw data used IDs for races/constructors; built lookup tables to map to readable names
  - Driver-team mapping: No table linked drivers to teams by season; reconstructed by grouping raw results data by driver, constructor, and season
  - Cross-dataset joins: wins, poles, and championships each required filtering results data and merging across multiple CSVs

Average Start Rate by Team × Debut Decade (Heatmap)

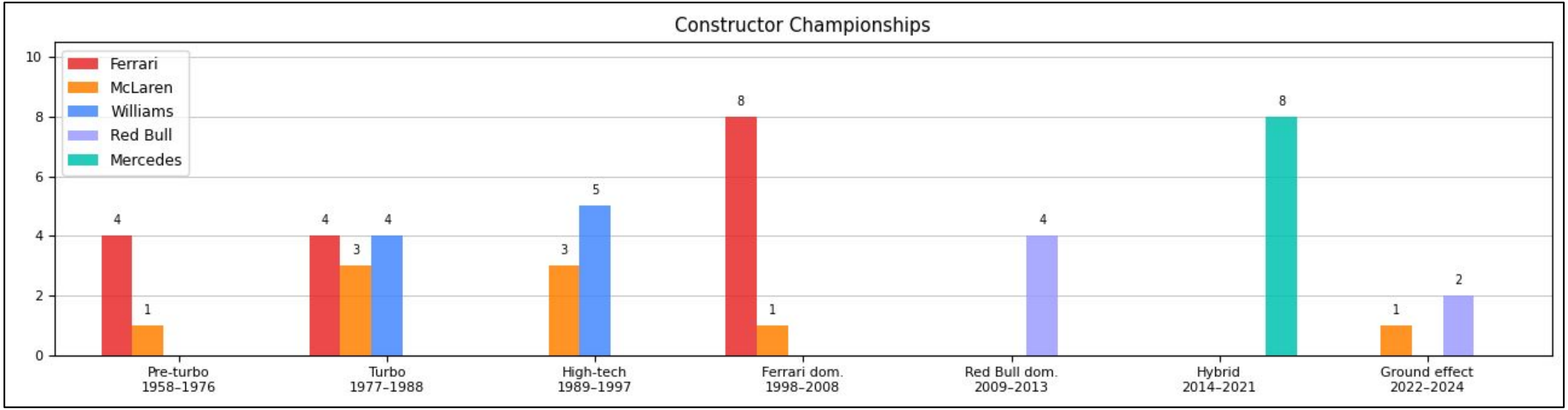


Average Start Rate by Season - Top 5 Long-Standing Teams (30+ Years)

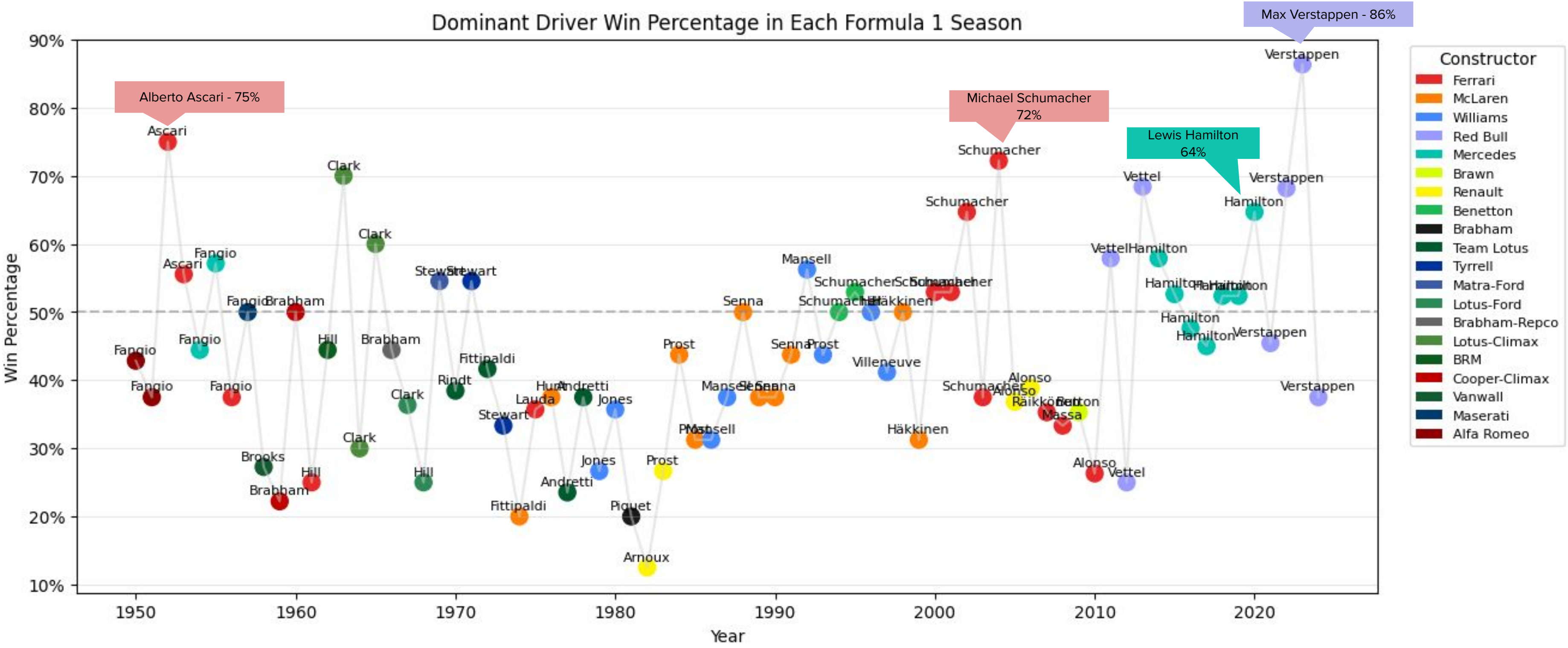



F1 Constructor Performance by Era
Race Wins
Ferrari
McLaren
Williams
Red Bull
Mercedes
0
20
40
60
80
100
39
17
30
49
40
19
37
63
96
55
10
12
20
1
47
4
17
1
28
111
10
6
47
5
Pre-turbo 1958–1976
Turbo 1977–1988
High-tech 1989–1997
Ferrari dom. 1998–2008
Red Bull dom. 2009–2013
Hybrid 2014–2021
Ground effect 2022–2024
Pole Positions
Ferrari
McLaren
Williams
Red Bull
Mercedes
0
20
40
60
80
100
120
58
12
24
28
29
14
40
79
82
61
17
4
14
2
57
9
23
1
1
16
118
23
8
30
7
Pre-turbo 1958–1976
Turbo 1977–1988
High-tech 1989–1997
Ferrari dom. 1998–2008
Red Bull dom. 2009–2013
Hybrid 2014–2021
Ground effect 2022–2024

F1 Constructor Performance by Era Contd.



Dominant Driver Win % in Each Formula 1 Season



Assumptions & Limitations

Dataset & Sources The analysis draws from multiple F1 CSVs assumed to be accurate and complete. Any errors in the source data carry through to the findings. Pole position data was pulled from two tables, with the qualifying table taking priority; gaps in older records may cause undercounting for early-era drivers.

Era Definitions The six F1 eras used in constructor analysis were sourced externally and applied as fixed date ranges. Technological transitions were gradual in reality, so these boundaries are approximations.

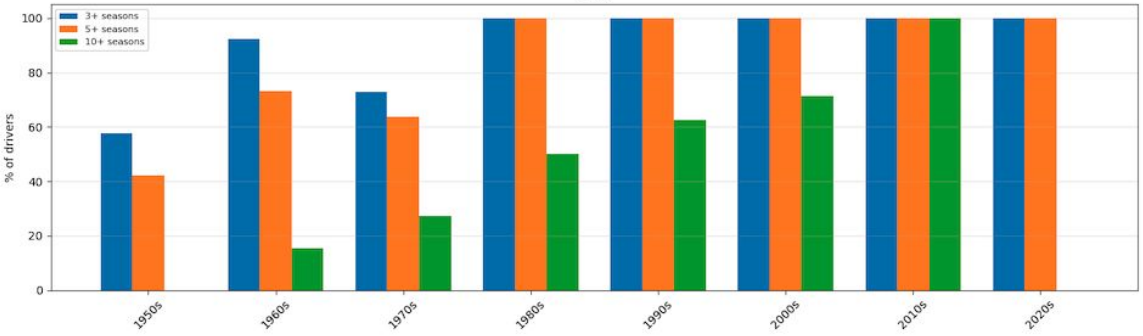
Car vs. Driver Performance Win rates and pole rates reflect both driver skill and car quality. This analysis does not separate the two, which is a fundamental limitation when comparing across eras.

Start Rate & DNQ Metrics Pre-qualifying in the 1980s and early 1990s heavily skews entries-vs-starts gaps for backmarker teams like Coloni, AGS, and Osella. Additionally, the Indianapolis 500 counted as a championship round in the 1950s-60s, inflating never-started counts for that period. The DNQ analysis is also restricted to teams with 15 or more drivers, so smaller historic entries are excluded.

Conclusion

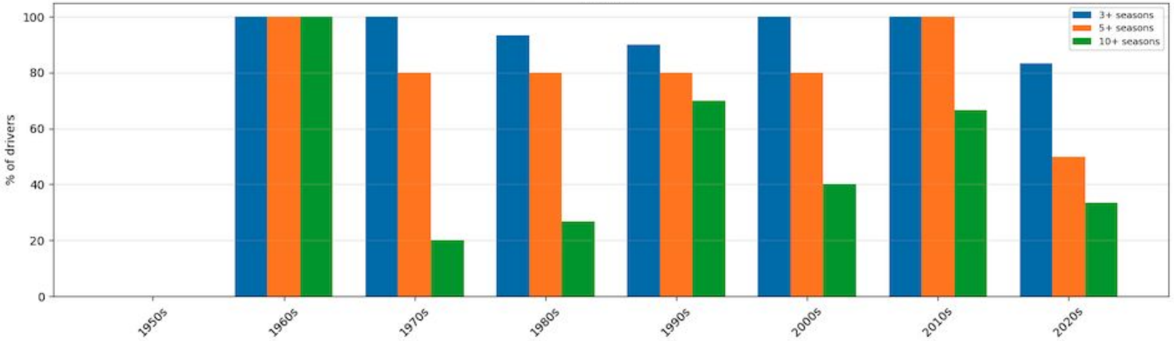
Driver Survival Analysis by Team and Debut Decade

Ferrari

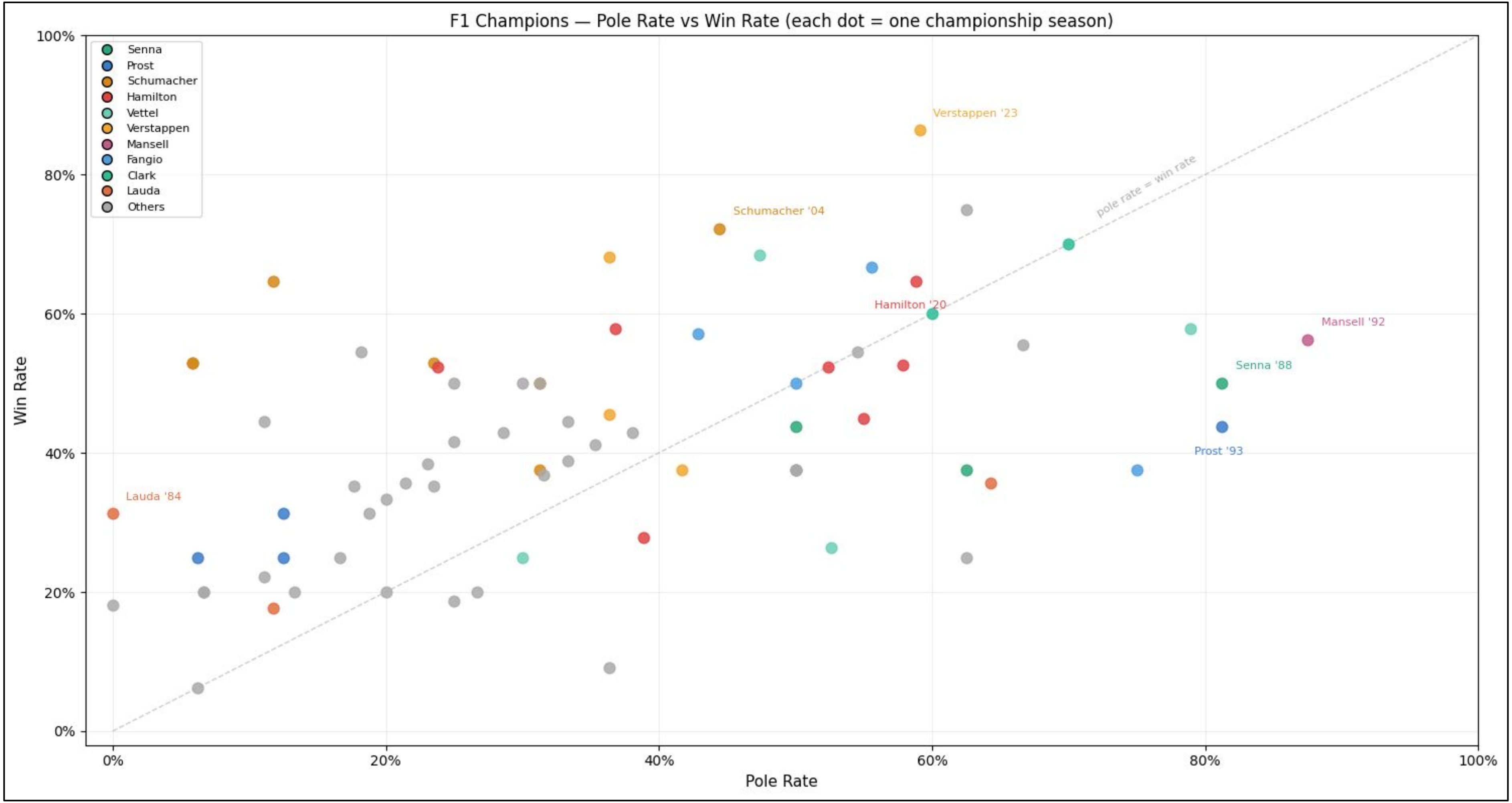


- Consistent growth over time
- Invest in a solid core

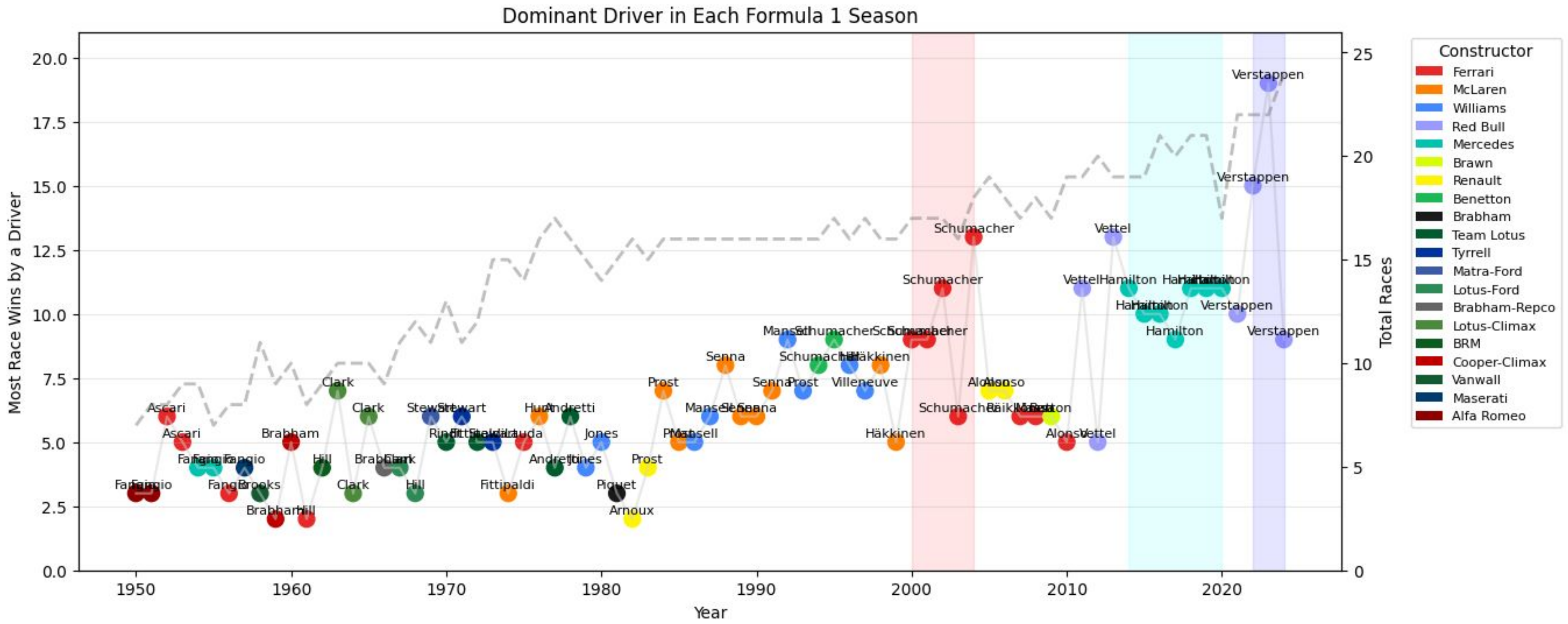
McLaren

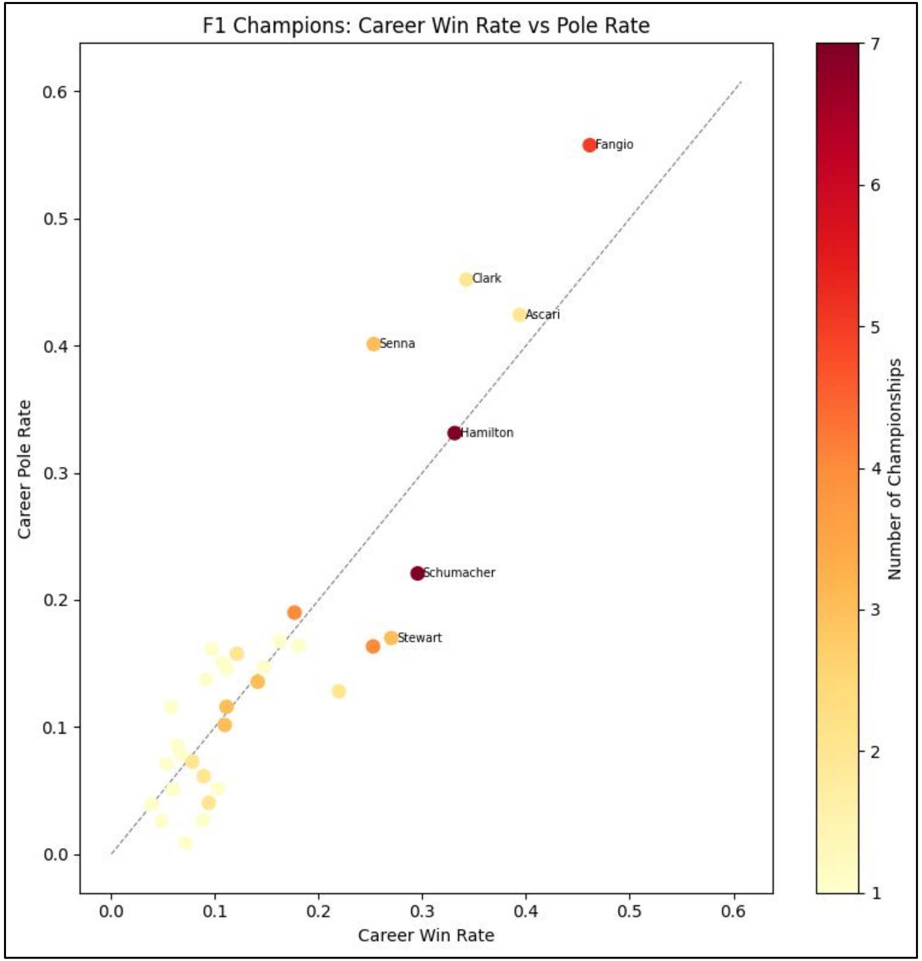


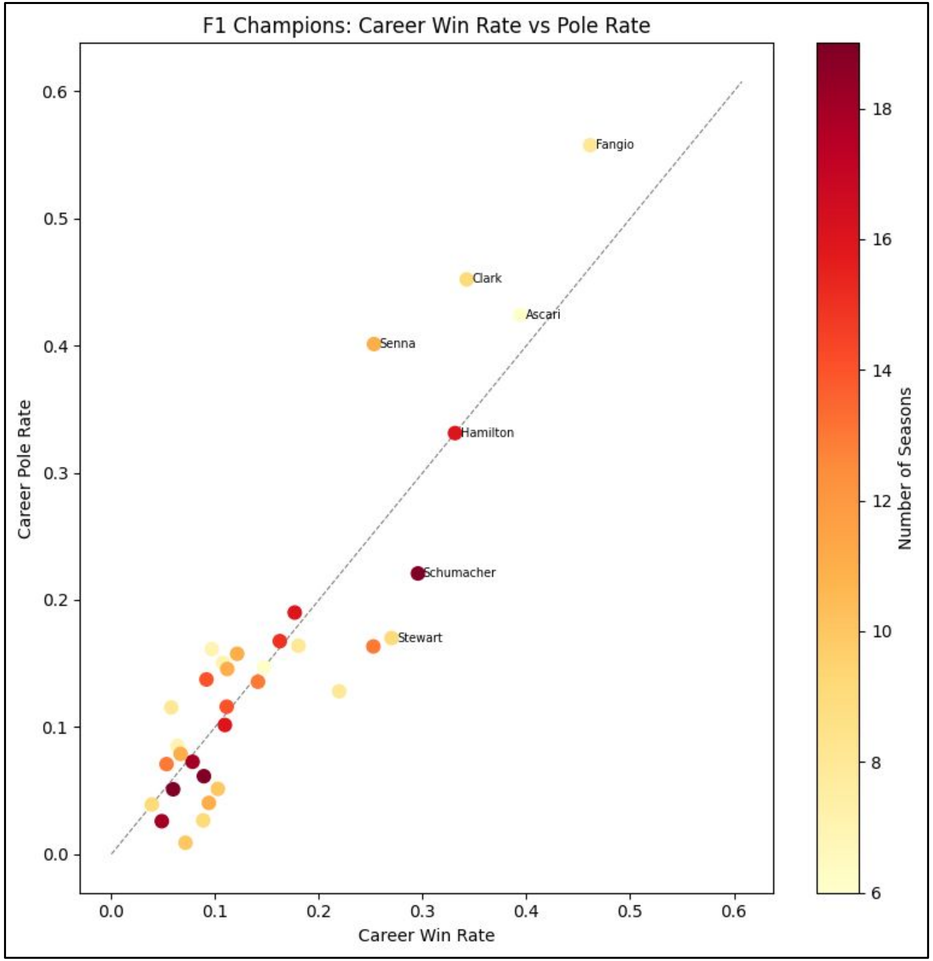
- Similar pattern to Ferrari until 1990s
- Cycle out drivers much quicker


F1 Champions — Pole Rate vs Win Rate (each dot = one championship season)
Senna
Prost
Schumacher
Hamilton
Vettel
Verstappen
Mansell
Fangio
Clark
Lauda
Others
Verstappen '23
Schumacher '04
Hamilton '20
Mansell '92
Senna '88
Prost '93
Lauda '84
pole rate = win rate
Win Rate
Pole Rate
0%
20%
40%
60%
80%
100%

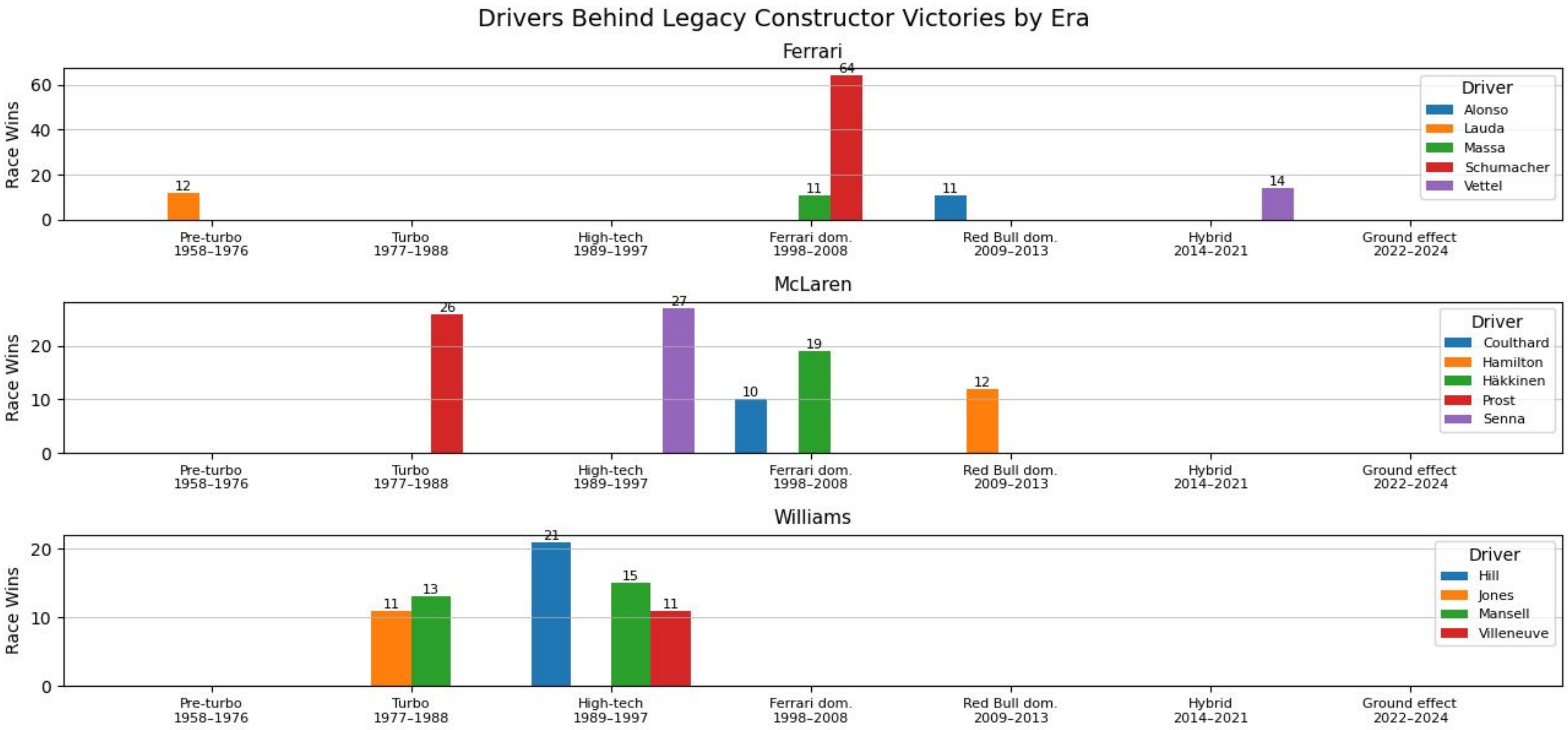
Dominant Driver in Each Formula 1 Season




F1 Champions: Career Win Rate vs Pole Rate
Career Pole Rate
Career Win Rate
Number of Championships
Fangio
Clark
Ascari
Senna
Hamilton
Schumacher
Stewart


F1 Champions: Career Win Rate vs Pole Rate
Career Pole Rate
Career Win Rate
Number of Seasons
Fangio
Clark
Ascari
Senna
Hamilton
Schumacher
Stewart

Drivers Behind Legacy Constructor Victories by Era



Drivers Behind Legacy Constructor Victories by Era Cont.